

Discrete – Problem 1.2.56

ee25btech11056 – Suraj.N

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Problem Statement

The iteration $x_{n+1} = \frac{1}{1+x_n^2}$ converges to a real number x in the interval $(0, 1)$ with $x_0 = 0.5$. The value of x correct up to 2 decimal places is:

- ① a) 0.65
- ② b) 0.68
- ③ c) 0.73
- ④ d) 0.80

Key Concept: Fixed-Point Iteration

A fixed point of a function $g(x)$ is a value x^* such that

$$g(x^*) = x^*.$$

The iteration $x_{n+1} = g(x_n)$ converges to x^* if the contraction condition holds:

$$|g'(x^*)| < 1.$$

Here $g(x) = \frac{1}{1+x^2}$, so we seek $x^* \in (0, 1)$ satisfying

$$x^* = \frac{1}{1+(x^*)^2}.$$

Applying MVT to the Iteration Error

Define the **error** at step n :

$$e_n = x_n - x^*$$

This measures how far you currently are from the answer.

Apply MVT to g between x_n and x^* :

$$g(x_n) - g(x^*) = g'(c) \cdot (x_n - x^*) \quad \text{for some } c \text{ between } x_n \text{ and } x^*$$

Now substitute what you know:

- ① Left side: $g(x_n) = x_{n+1}$ and $g(x^*) = x^*$, so left side = e_{n+1}
- ② Right side: $(x_n - x^*) = e_n$

This gives the key result:

$$e_{n+1} = g'(c) \cdot e_n$$

The error at the next step equals the current error multiplied by $g'(c)$.

Therefore, $|g'(x^*)| < 1 \Rightarrow$ error shrinks every step $\Rightarrow$ **converges**

Setting Up the Fixed-Point Equation

Multiply both sides by $1 + (x^*)^2$:

$$x^*(1 + (x^*)^2) = 1.$$

Expanding:

$$(x^*)^3 + x^* - 1 = 0.$$

Let $f(x) = x^3 + x - 1$. We locate the root:

① $f(0) = -1 < 0$

② $f(1) = 1 > 0$

By the Intermediate Value Theorem, a root exists in $(0, 1)$.

Verifying the Contraction Condition

Compute the derivative of $g(x)$:

$$g'(x) = \frac{-2x}{(1+x^2)^2}.$$

At $x^* \approx 0.68$:

$$g'(0.68) = \frac{-2(0.68)}{(1+(0.68)^2)^2} = \frac{-1.36}{(1.4624)^2} \approx \frac{-1.36}{2.139} \approx -0.636.$$

Since $|g'(x^*)| \approx 0.636 < 1$, the contraction condition is satisfied and the iteration **converges**.

Solving the given equation

Solve:

$$x^3 + x - 1 = 0$$

This is in standard cubic form:

$$x^3 + px + q = 0$$

Comparing:

$$p = 1, \quad q = -1$$

Using Cardano formula for equation:

$$x^3 + px + q = 0$$

Solution:

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{D}} + \sqrt[3]{-\frac{q}{2} - \sqrt{D}}$$

where discriminant:

$$D = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3$$

Calculating the discriminant

$$D = \left(\frac{-1}{2}\right)^2 + \left(\frac{1}{3}\right)^3$$

$$D = \frac{1}{4} + \frac{1}{27}$$

$$D = \frac{27 + 4}{108} = \frac{31}{108}$$

Substituting the value of the discriminant to find x :

$$x = \sqrt[3]{\frac{1}{2} + \sqrt{\frac{31}{108}}} + \sqrt[3]{\frac{1}{2} - \sqrt{\frac{31}{108}}}$$

$$x = \sqrt[3]{0.5 + 0.536} + \sqrt[3]{0.5 - 0.536}$$

$$x = \sqrt[3]{1.036} + \sqrt[3]{-0.036}$$

$$x \approx 1.012 - 0.33 = 0.68$$

Conclusion

The fixed-point iteration $x_{n+1} = \frac{1}{1+x_n^2}$ with $x_0 = 0.5$ converges to the unique root of

$$x^3 + x - 1 = 0 \quad \text{in } (0, 1).$$

The converged value correct to 2 decimal places is

$$x^* \approx 0.68.$$

The correct answer is **b) 0.68**.

Plot of Sequence

Convergence of $x_{n+1} = \frac{1}{1+x_n^2}$, $x_0 = 0.5$

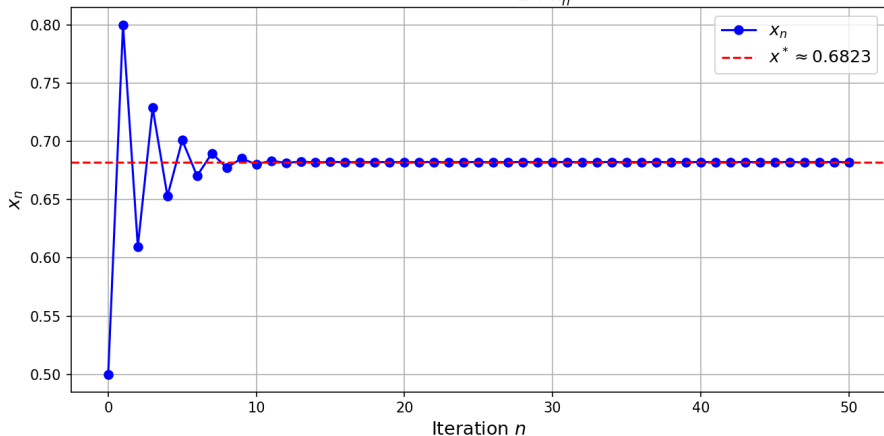


Figure: Sequence

Python Code: Convergence Value

```
def g(x):
    return 1.0 / (1.0 + x**2)

x = 0.5
tol = 1e-10
max_iter = 1000
x_new = x

for i in range(max_iter):
    x_new = g(x)
    print(f"The_{i}_iteration_value_is_{x_new}")
    if abs(x_new - x) < tol:
        print(f"Converged_at_iteration_{i+1}")
        break
    x = x_new

print(f"Fixed_point_x*_{x_new:.10f}")
print(f"Rounded_to_2_decimal_places_{round(x_new,2)}")

# Verification: solve  $x^3 + x - 1 = 0$ 
import numpy as np
roots = np.roots([1, 0, 1, -1])
real_roots = [r.real for r in roots if abs(r.imag) < 1e-10]
print(f"Root_of_ $x^3+x-1=0$  in (0,1) :_{real_roots[0]:.10f}")
```

Python Code: Convergence Plot

```
import numpy as np
import matplotlib.pyplot as plt

def g(x):
    return 1.0 / (1.0 + x**2)
x = 0.5
iterates = [x]
tol = 1e-10
max_iter = 50
for i in range(max_iter):
    x_new = g(x)
    iterates.append(x_new)
    if abs(x_new - x) < tol:
        print(f"Converged at iteration {i+1}")
        break
    x = x_new
x_star = iterates[-1]
n_vals = list(range(len(iterates)))

plt.figure(figsize=(9, 5))
plt.plot(n_vals, iterates, 'bo-', markersize=6,
         linewidth=1.5, label=r'$x_n$')
plt.axhline(y=x_star, color='r', linestyle='--',
            linewidth=1.5, label=f'$x_{\star}$ {x_star:.4f}')
plt.xlabel('Iterationn', fontsize=13)
plt.ylabel('$x_n$', fontsize=13)
plt.title('Convergence of Fixed-Point Iteration', fontsize=13)
plt.legend(fontsize=12)
plt.grid(True)
plt.tight_layout()
plt.savefig('convergence_plot.png', dpi=150)
plt.show()
```